MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548148

(CONTROL, SEARCH/TRACK, AND GAIN)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification, together with the general requirements of MIL-C-50652, comprise the complete requirements for the search/track, and gain control assembly used in Range Computer, CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10548148	Assembly, Control, Search/Track and Gain

PACKING DATA SHEET

10548148	Packaging of Search/Track and Gain Control

FSC 1285

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548148 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - For the requirements of 3.5.3 through 3.5.4 logic input and output levels shall be as follows:

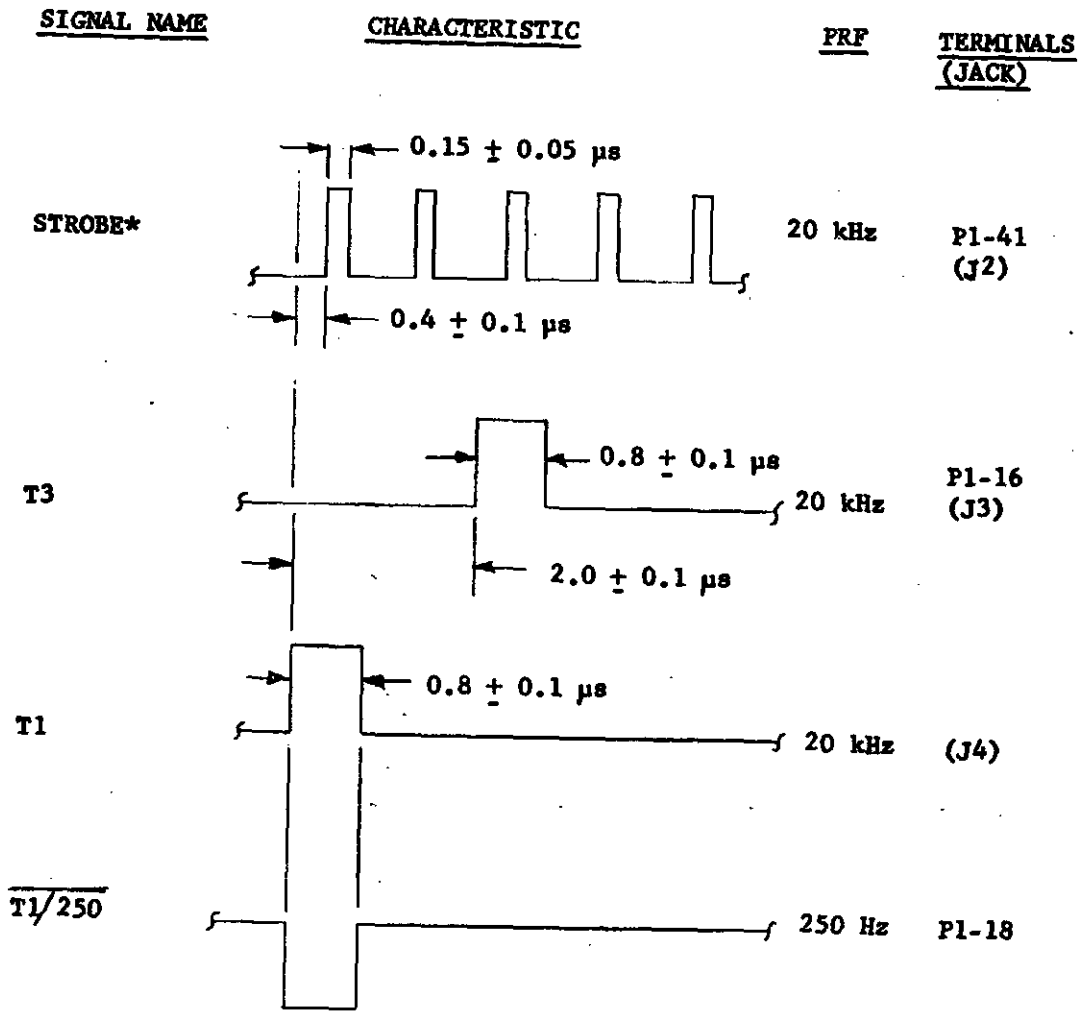
Logic 0: 0.0 ±0.5 volt direct current (Vdc) level
Logic 1: +3.5 ±1.0 Vdc level

3.5.1 Input power. - The input power applied between terminals P1-30, 31 and 32 (+5 V) and terminals P1-24, 25 and 26 (+5 V return) shall be 5.0 ±0.3 Vdc, at 150 milliamperes (mA), regulated, with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signals. - Input signals to the search/track and gain control assembly shall be STROBE, T3 and $\overline{T1/250}$ as shown in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ±0.5 V to +3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V.

3.5.3 "TRACK" signal. - With the logic level conditions, for P1-14, P1-20, P1-1, and P1-44, sequentially established as specified in Table I, Steps 1 through 11, the track signal at P1-47 shall have the logic levels as specified in Table I. When the logic levels for P1-5 and P1-7 are simultaneously changed the track output level shall change with the delays as specified in Table I.

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* STROBE signal is a train of 15 pulses at a 20 kHz PRF.

FIGURE 1. INPUT SIGNAL CHARACTERISTICS.

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TABLE I. - INPUT-OUTPUT SIGNALS

STEP	LOGIC LEVEL INPUT CONDITIONS							OUTPUT SIGNAL DELAY
	P1-14 (S1)	P1-20 (S2)	P1-1 (S3)	P1-44 (S5)	P1-7	P1-5 (S4)	P1-47 (TP3)	
1	0	1	1	1	1	0	0	-
2					0	1	1	None
3					1	0	0	0.80 to 1.20 s
4	1	1	1	1	0	1	1	-
5					1	0	0	2.75 to 3.25 s
6	1	0	1	1	0	1	1	-
7					1	0	0	None
8	1	0	0	1	1	0	1	None
9	1	0	1	1	1	0	0	None
10	1	0	1	1	0	1	1	None
11	1	0	0	1	0	1	1	None
12	1	1	1	0	1	0	Fig. 2	

3.5.3.1 TRACK signal test. - With the input logic levels as specified in Table I, Step 12, the track output signal shall be as specified in Figure 2.

3.5.4 Gain signals. - For 3.5.4.1 through 3.5.4.3 the input logic levels for P1-20, P1-14, P1-1, and P1-44 shall be as specified for Table I, Step 1.

3.5.4.1 G1 signal. - The G1 output signal at P1-11 shall change from logic 1 to logic 0 between 0.125 and 0.375 s after the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table II, Step 1 to Step 2. The G1 signal shall change from logic 0 to logic 1 between 0.80 and 1.20 s when the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table II, Step 2 to Step 1.

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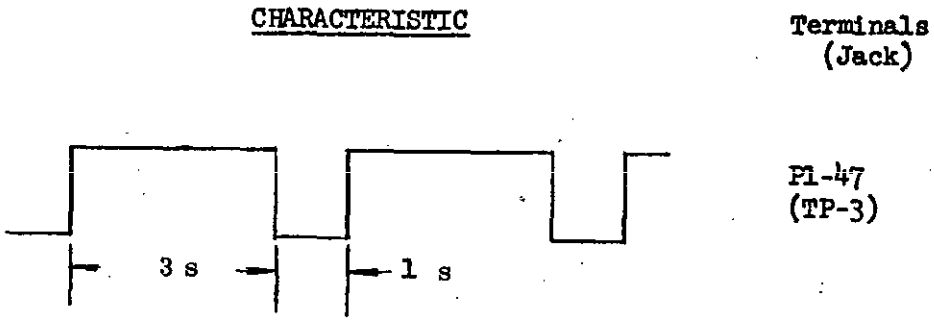


FIGURE 2. OUTPUT SIGNAL CHARACTERISTIC.

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TABLE II. - INPUT SIGNALS

INPUT LOGIC LEVELS

STEP	P1-7	P1-5 (S4)
1	1	0
2	0	1

3.5.4.2 G2 signal. - The G2 output signal at P1-15 shall be a single positive pulse between 0.125 and 0.375 s wide which occurs when the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table III, Step 1 to Step 2.

3.5.4.3 G3 signal. - The G3 output signal at P1-13 shall change from logic 0 to logic 1 between 0.25 and 0.75 s after the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table III, Step 1 to Step 2. The G3 signal shall change from logic 1 to logic 0 immediately when the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table III, Step 2 to Step 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special samplings. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

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4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table III shall be utilized in the performance of the applicable tests as specified in 4.6.3.

TABLE III. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 250 mA.
2. Pulse generator	Train of 15 pulses at 20 kHz externally synchronized. Amplitude 3.5 ±1.0 V peak-to-peak into 50-ohm load.
3. Pulse generator	Two separate output pulses, 20 kHz PRF, controllable in separation, amplitude 3.5 ±1.0 V peak-to-peak into 50-ohm load.
4. Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less, sweep period 50 ns per cm or greater, input resistance greater than 8 megohms, input capacitance less than 25 PF.
5. Test fixture	As specified in Figure 3. See Table IV for test fixture parts list.

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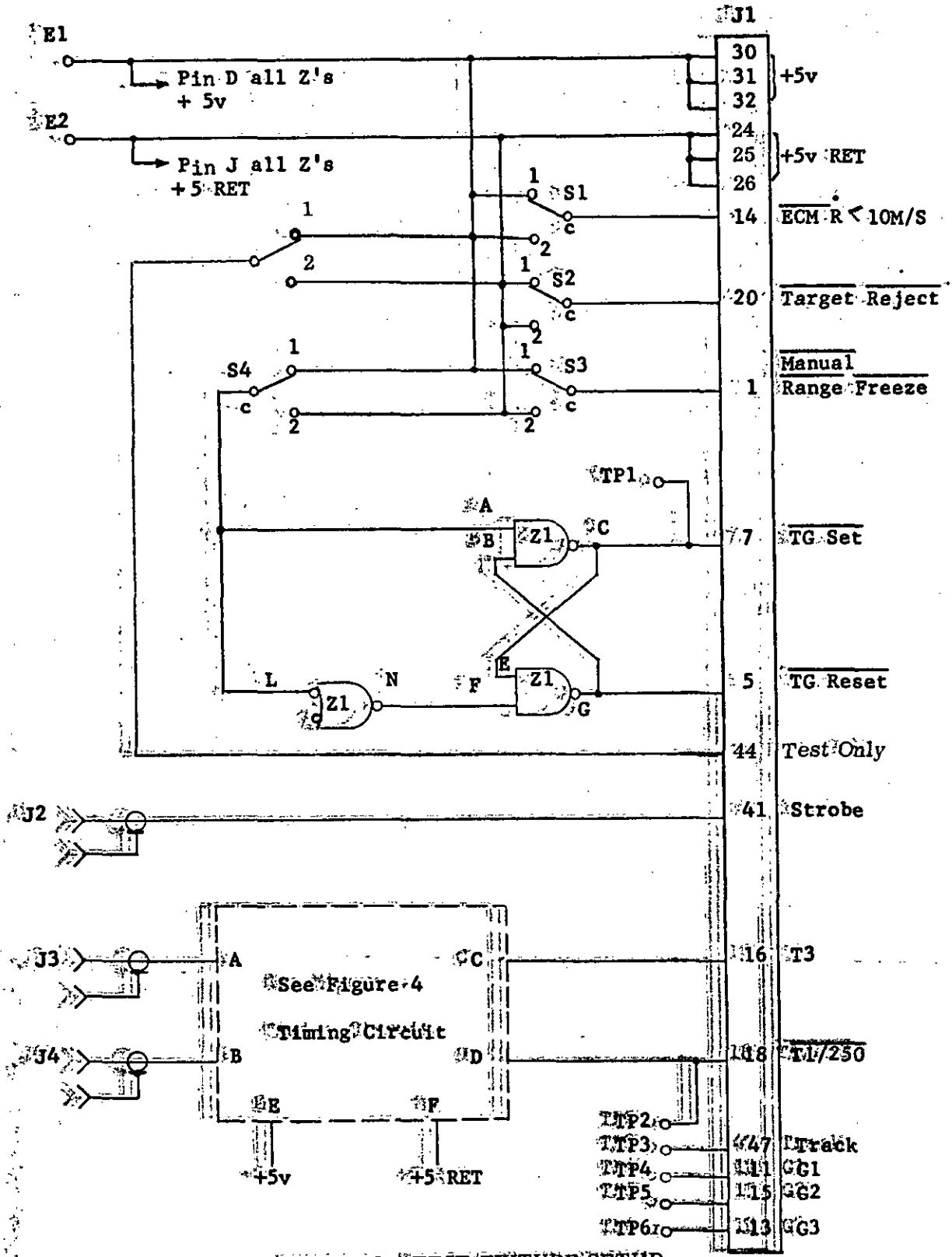


FIGURE 3. TEST FIXTURE SETUP.

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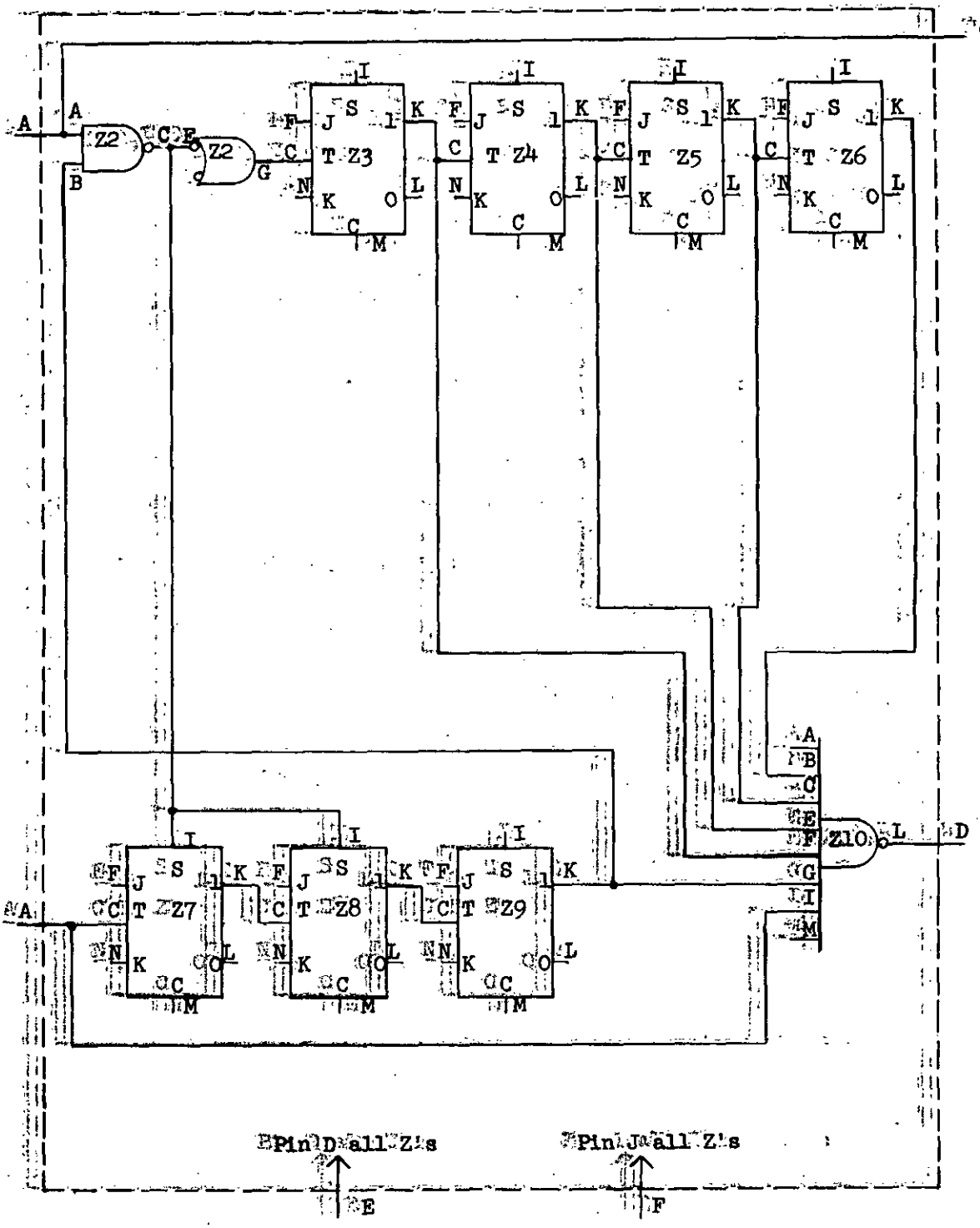


FIGURE 4. TEST FIXTURE TIMING CIRCUIT.

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TABLE IV. - TEST FIXTURE PARTS LIST

DESIGNATION	
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector, any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, *UG-25B/U or equivalent.*
J3	Same as J2.
J4	Same as J2.
S1 thru S5	Switch SPDT.
TP1 thru TP6	Tip jack, H. H. Smith 1501 or equivalent.
Z1 and Z2	Integrated circuit, SG-140-03, Sylvania or equivalent.
Z3 thru Z9	Integrated circuit, SF50-03, Sylvania or equivalent.
Z10	Integrated circuit, SG60-21, Sylvania or equivalent.

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4.6. Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - Set up equipment as shown in Figure 5. Use oscilloscope, item 4, for observation of input and output signals specified in 4.6.3.2 through 4.6.3.4. For observation of delay times use oscilloscope, item 4, in single sweep mode of operation using TP1 trigger for synchronization. It will be necessary to reset the oscilloscope after each delay test step is initiated. Switches S1 through S3 and S5, on the test fixture, item 5, are used to supply the logic level input signals specified in Table I. These switches supply a logic 1 in position 1 and a logic 0 in position 2. Switch S5 is used to supply the logic levels for Tables I and II. This switch supplies the logic levels of Table II, step 1, when in position 2 and the levels of step 2 when in position 1.

4.6.3.1 Input power. - Using power supply, item 1, apply input power as specified in 3.5.1.

4.6.3.2 Input signals. - Use pulse generators, items 2 and item 3, to supply the strobe signals T3 and T1, as shown of Figure 1. Connect these signals to J2, J3 and J4 respectively. At TP2, check the T1/250 input signal for the requirements of Figure 1.

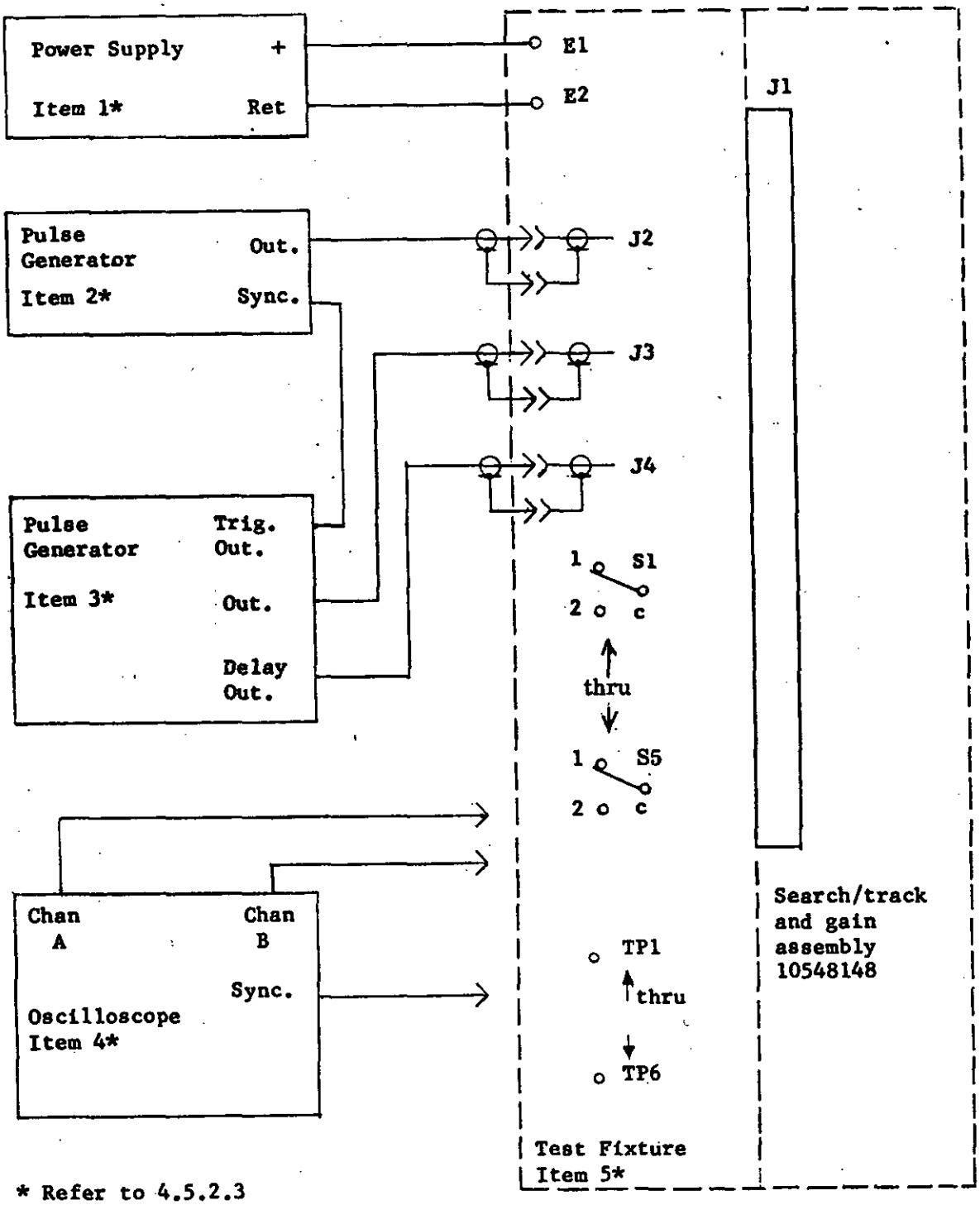
4.6.3.3 TRACK signal. - Using switches S1 through S5 to establish the logic levels as specified in 3.5.3, observe, at TP3, the track output signal as specified in 3.5.3.

4.6.3.3.1 TRACK signal test. - Using switches S1 through S5 to establish the logic levels of Table I, Step 12, observe, at TP3, the track signal as shown in Figure 2.

4.6.3.4 Gain signals. - Using switches S1 through S3 and S5 establish the logic level conditions of Table I, Step 1.

4.6.3.4.1 G1 signal. - While using switch S4 to establish the logic level conditions of Table II, observe at TP4, the G1 signal as specified in 3.5.4.1.

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* Refer to 4.5.2.3

FIGURE 5. SEARCH/TRACK AND GAIN CONTROL TEST SETUP.

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4.6.3.4.2 G2 signal. - While using switch S4 to establish the logic level conditions of Table II, observe, at TP5, the G2 signal as specified in 3.5.4.2.

4.6.3.4.3 G3 signal. - While using switch S4 to establish the logic level conditions of Table II, observe at TP6, the G3 signal as specified in 3.5.4.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548148. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The search/track and gain control assembly is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The search/track and gain control generates three gain control signals to be used in the range computer upon initial target acquisition.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

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Preparing Activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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